

AI-Powered Interior Design Platform Architecture & Development Roadmap

Phase 1 - Application Foundation

Build the main application using React (frontend), FastAPI (backend), PostgreSQL (database), and JWT authentication. The application manages user accounts, projects, design history, cloud storage, and project versions. All project data is stored through REST APIs exposed by FastAPI.

Phase 2 - 2D CAD Engine

Implement a CAD engine using TypeScript, HTML5 Canvas/SVG (or Three.js for web drawing) with geometry algorithms. Users can draw walls, rooms, doors, windows, stairs, dimensions, and layers. Object snapping, undo/redo, rotation, scaling, alignment, and measurement tools operate similarly to CAD software.

Phase 3 - AI Prompt Interpreter

Integrate an LLM API with Python/FastAPI. The user enters a natural language prompt describing budget, style, room count, colors, furniture preferences, and constraints. The AI converts the prompt into structured JSON used by the platform.

Phase 4 - AI Floor Planning Engine

Using computational geometry and optimization algorithms in Python, generate an optimized 2D floor plan from the parsed prompt. The engine allocates room sizes, circulation space, and layout based on available area and user constraints.

Phase 5 - AI Vastu Validation

A rule engine compares the generated floor plan against a Vastu knowledge base. If rooms violate rules, the engine automatically rearranges layouts while minimizing changes.

Phase 6 - AI Material Recommendation

The recommendation engine combines the user budget, preferred style, durability requirements, and available catalog data. Python services rank suitable flooring, paint, furniture, lighting, kitchen finishes, and decor.

Phase 7 - AI Furniture Placement & Auto Alignment

A spatial reasoning engine places furniture automatically. Constraint-solving and collision detection algorithms ensure proper walking space, door clearance, wall alignment, and proportional spacing. AI centers sofas with TVs, aligns dining tables, keeps beds against appropriate walls, and automatically rotates furniture to optimal angles based on room geometry and user requests.

Phase 8 - 3D Scene Generator

The approved 2D plan is converted into a 3D scene using Blender APIs or Three.js. Walls, floors, ceilings, doors, windows, and furniture are generated automatically from CAD geometry.

Phase 9 - AI Camera & Lighting

AI analyzes room dimensions and automatically selects camera positions, viewing angles, focal length, lighting placement, HDRI environment, exposure, and composition to produce visually balanced renders.

Phase 10 - Photorealistic Rendering

Use Blender Cycles or another rendering engine to generate realistic images. Physically Based Rendering (PBR), global illumination, reflections, shadows, and realistic materials are applied

automatically.

Phase 11 - Budget & Shopping Engine

The platform generates an itemized estimate, material quantities, labor estimates, and shopping recommendations. The report updates automatically whenever the design changes.

Phase 12 - Complete Workflow

Workflow: 1. User enters requirements. 2. AI interprets the prompt. 3. AI generates the floor plan. 4. AI validates Vastu. 5. AI recommends materials. 6. AI places and aligns furniture. 7. AI creates the 3D model. 8. AI optimizes camera angles and lighting. 9. AI renders realistic images. 10. AI generates the final budget and material report.